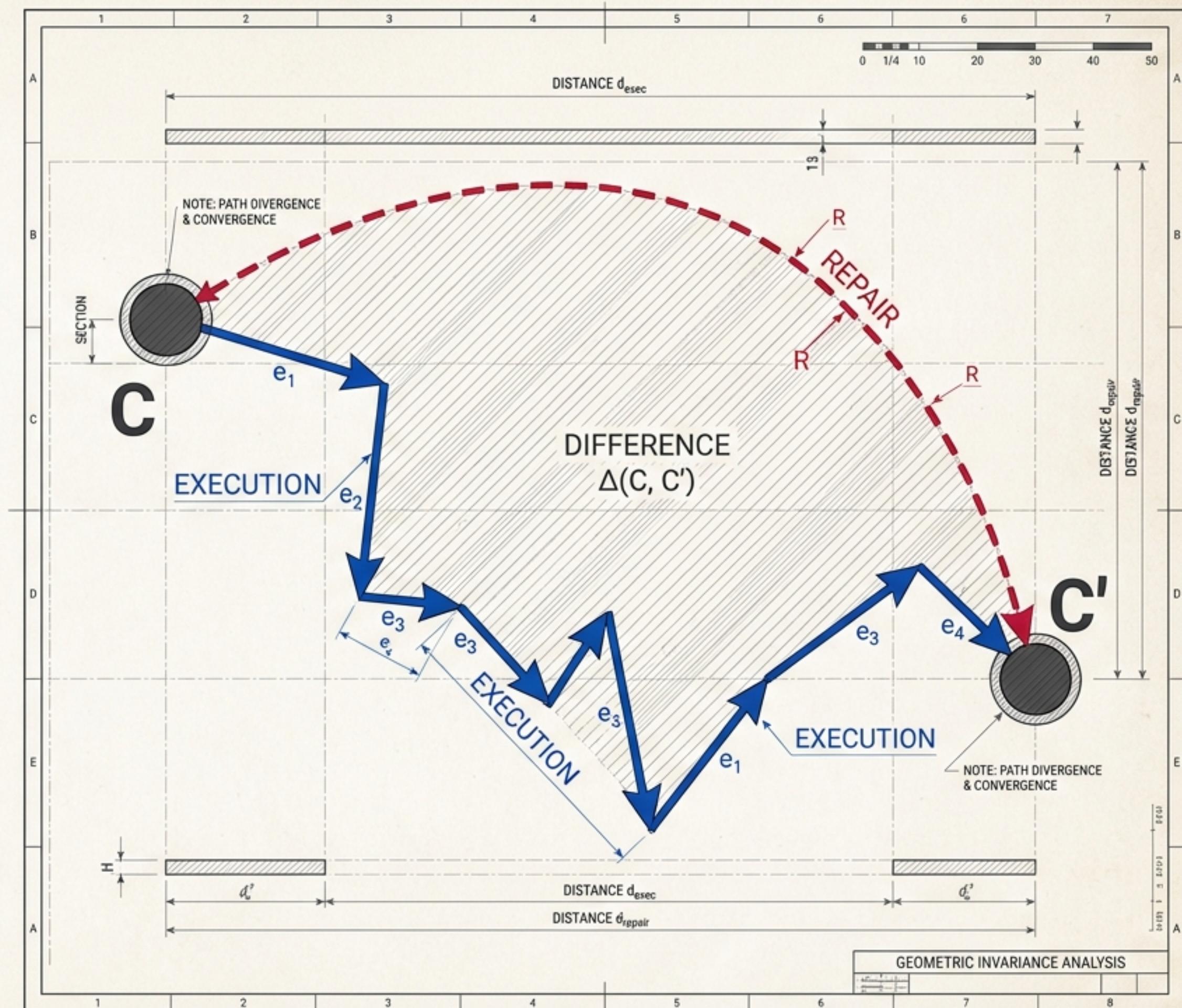


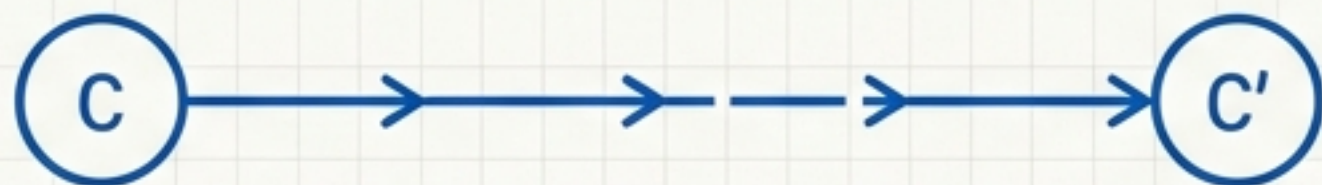
THE DIRECTED GRAPH & THE SHADOW

The Local
Geometry of
Continuation and the
Limits of Invariance



Every continuation system carries two independent primitives of proximity

Executable Cost (E)

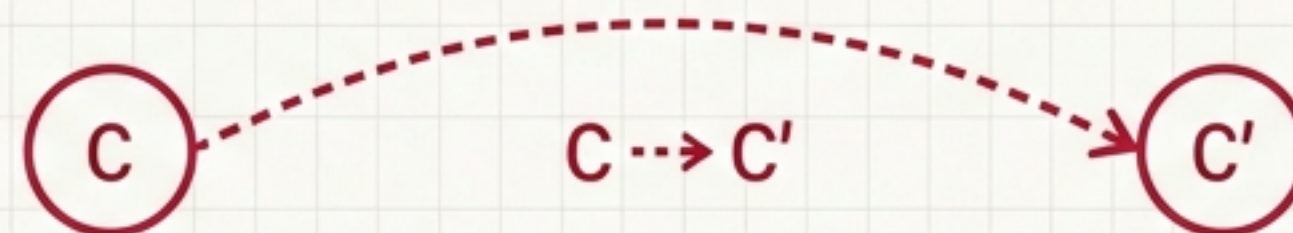


Origin: Generated internally by the system's own operational algebra.

Meaning: The absolute least total cost of a sequence of operations to reach C' from C .

Nature: Procedural. Driven by free enrichment of the operational graph.

Repair Contrast (R)

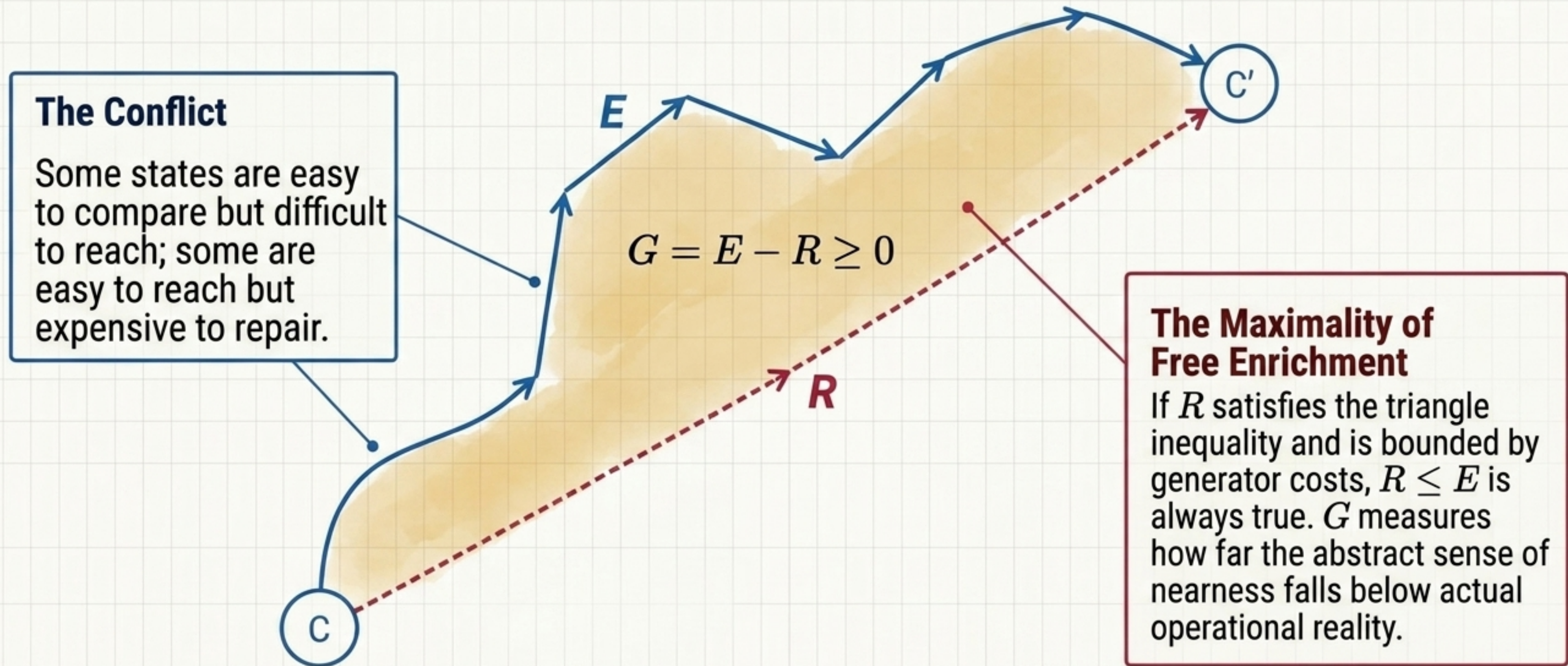


Origin: Imposed externally, independent of available operations.

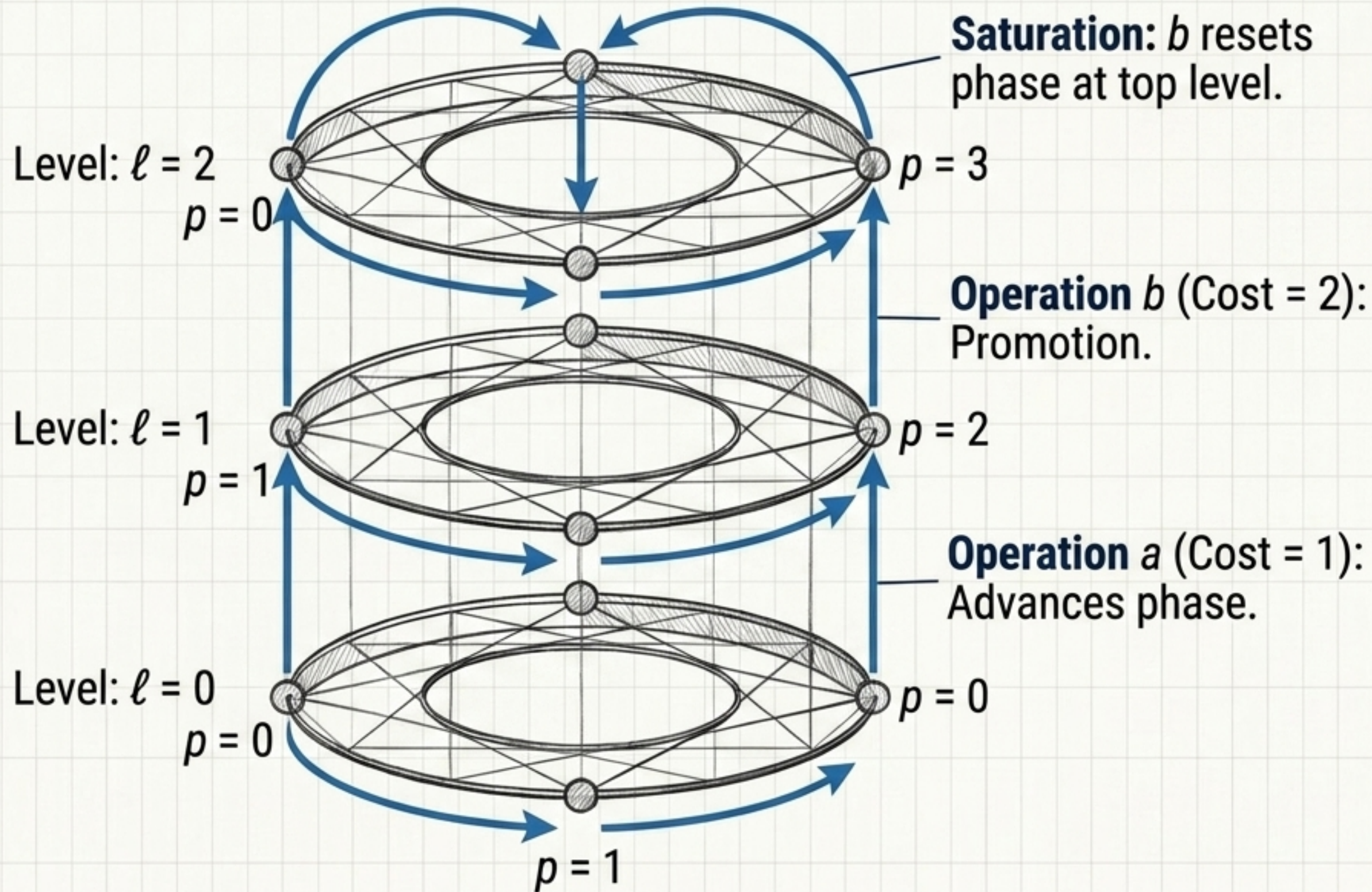
Meaning: The cost of recognizing C' as a repaired form of C .

Nature: Representational. Directed comparison.

The Coherence Gap: Representational nearness and operational accessibility dissociate



Grounding the theory: A fully explicit twelve-state continuation system



Context Note:

Every claim in the theory reduces to a computable table in this 12-state model.

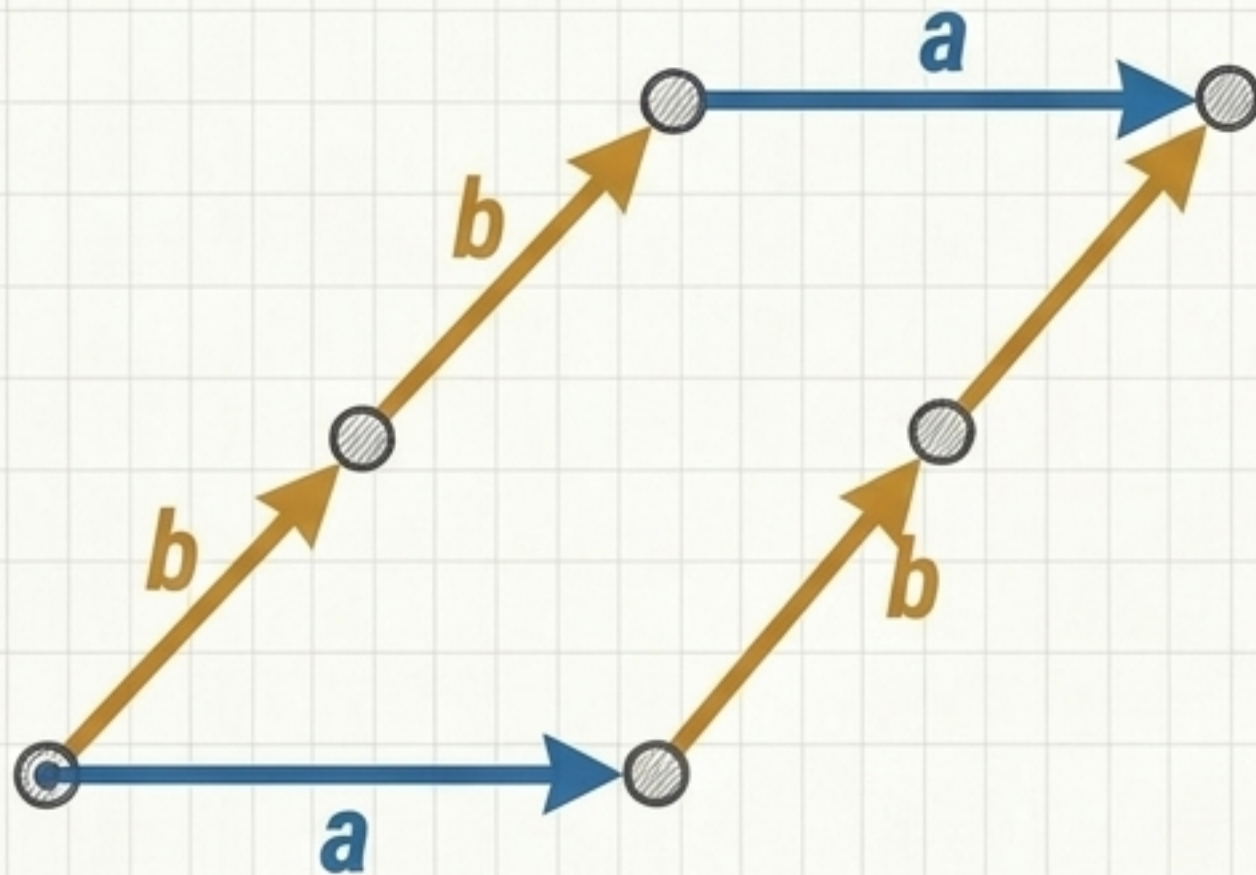
Recognizing an upgrade is cheap; recognizing a downgrade is expensive but finite; phase mismatch costs 0.5 per step.

Order dependence concentrates at the boundary of saturation

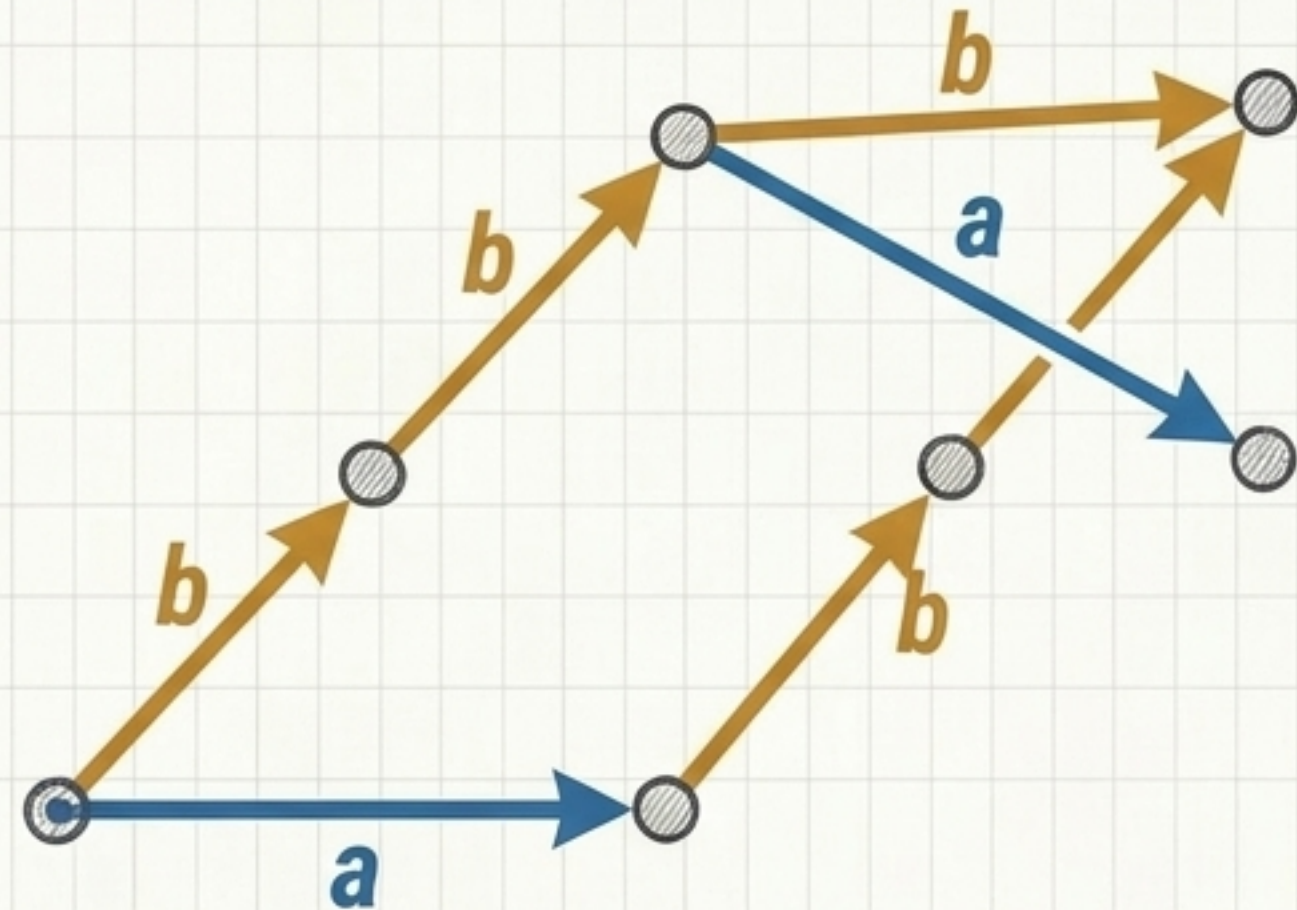
The Torsion Square

Figure A

Level 1 (Regular Stratum)



Level 2 (Singular Stratum)



The generators commute everywhere except on the singular stratum. This localized torsion records the failure of geometric transport to represent operational reality.

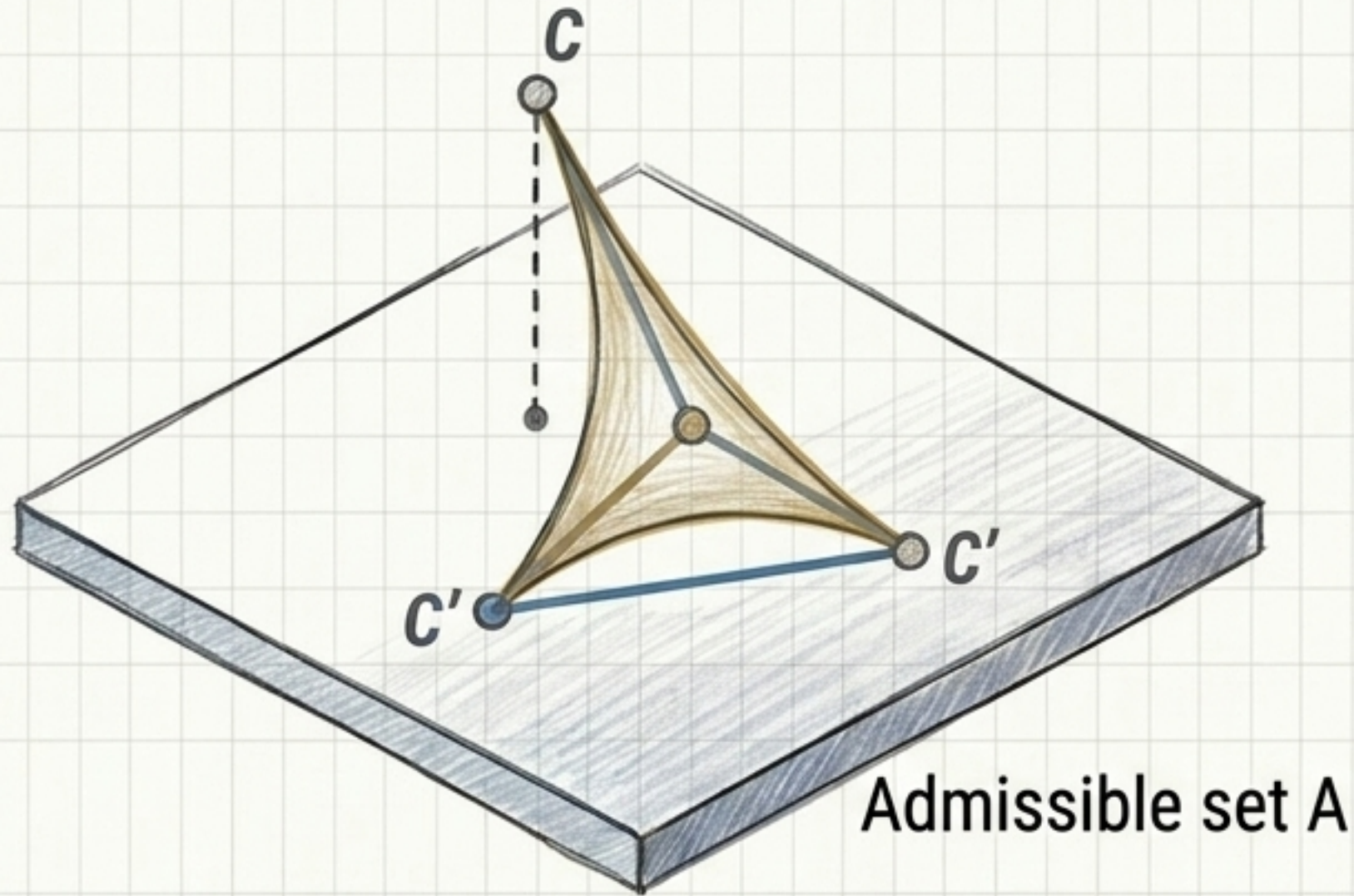
Repair structures bifurcate into two distinct geometrical regimes

Enriched Repair (Lawvere)	Divergence Repair (Information Geometry)
Yes.	No.
Local (verified generator by generator).	Global axiom (no local certificate exists).
Always ≤ 0 (projection can only lose).	Positive interference is possible.

“Positive Pythagorean interference — the signature of dually flat information geometry... is impossible in enriched repair. At the same location... enriched repair admits a local certificate of coherence... Repair structures therefore bifurcate...”

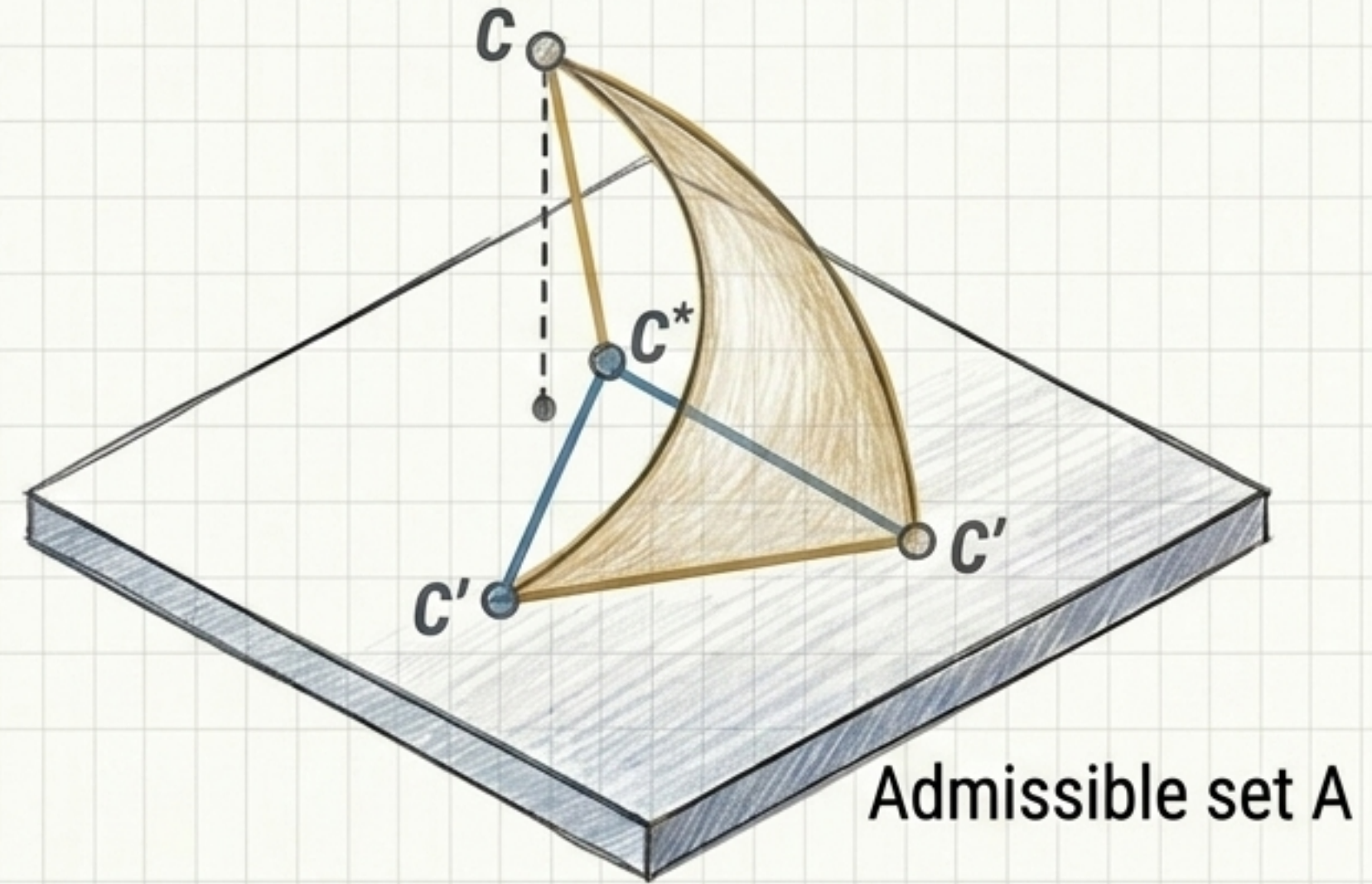
The Projection Defect separates regimes by an observable sign

Enriched / Lawvere Regime



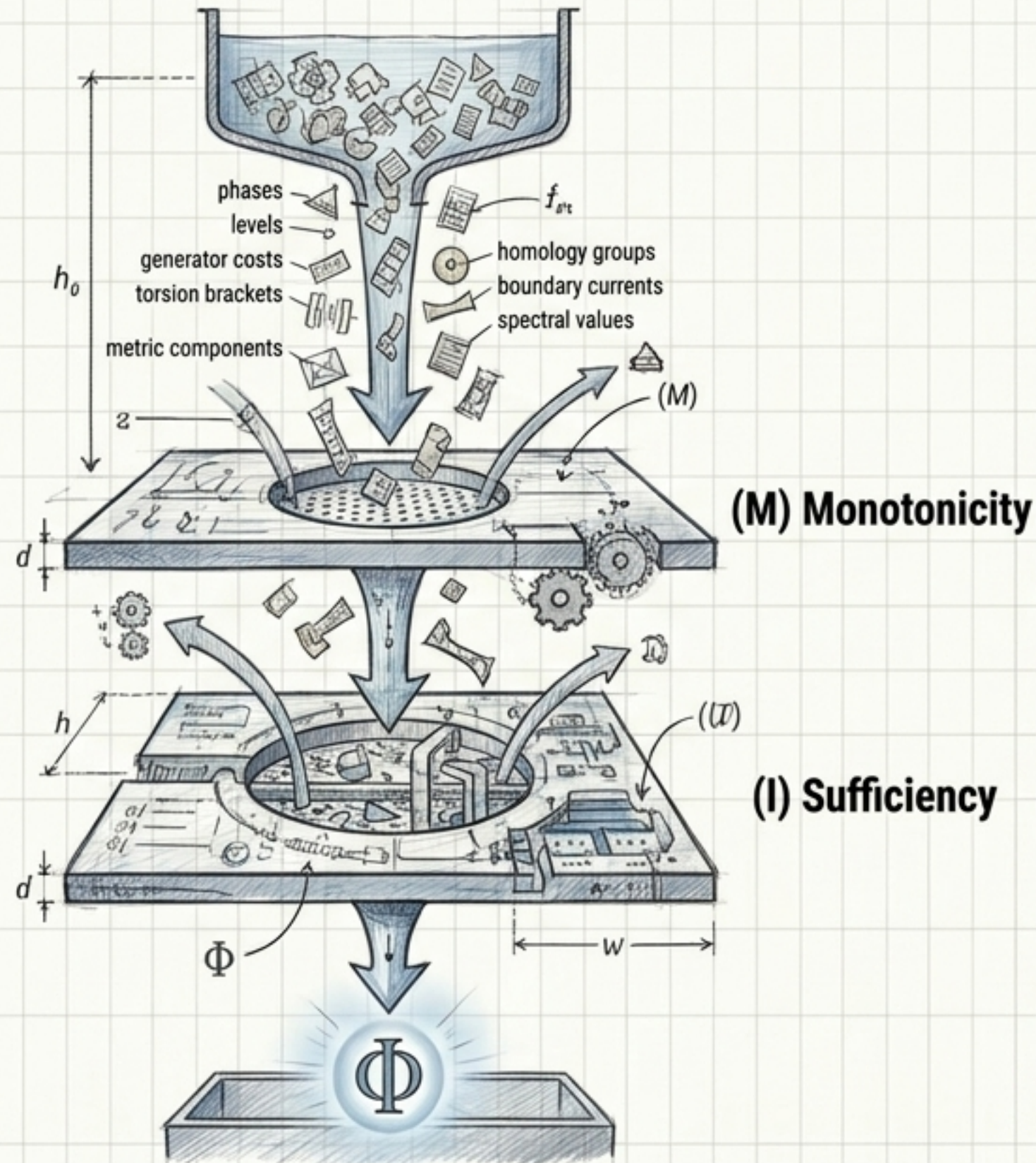
$\Delta_A \leq 0$. The triangle inequality holds.

Divergence Regime



$\Delta_A > 0$. The generalized Pythagorean theorem yields exact positive decompositions.

The Finite Characterization Problem: Searching for the true invariant



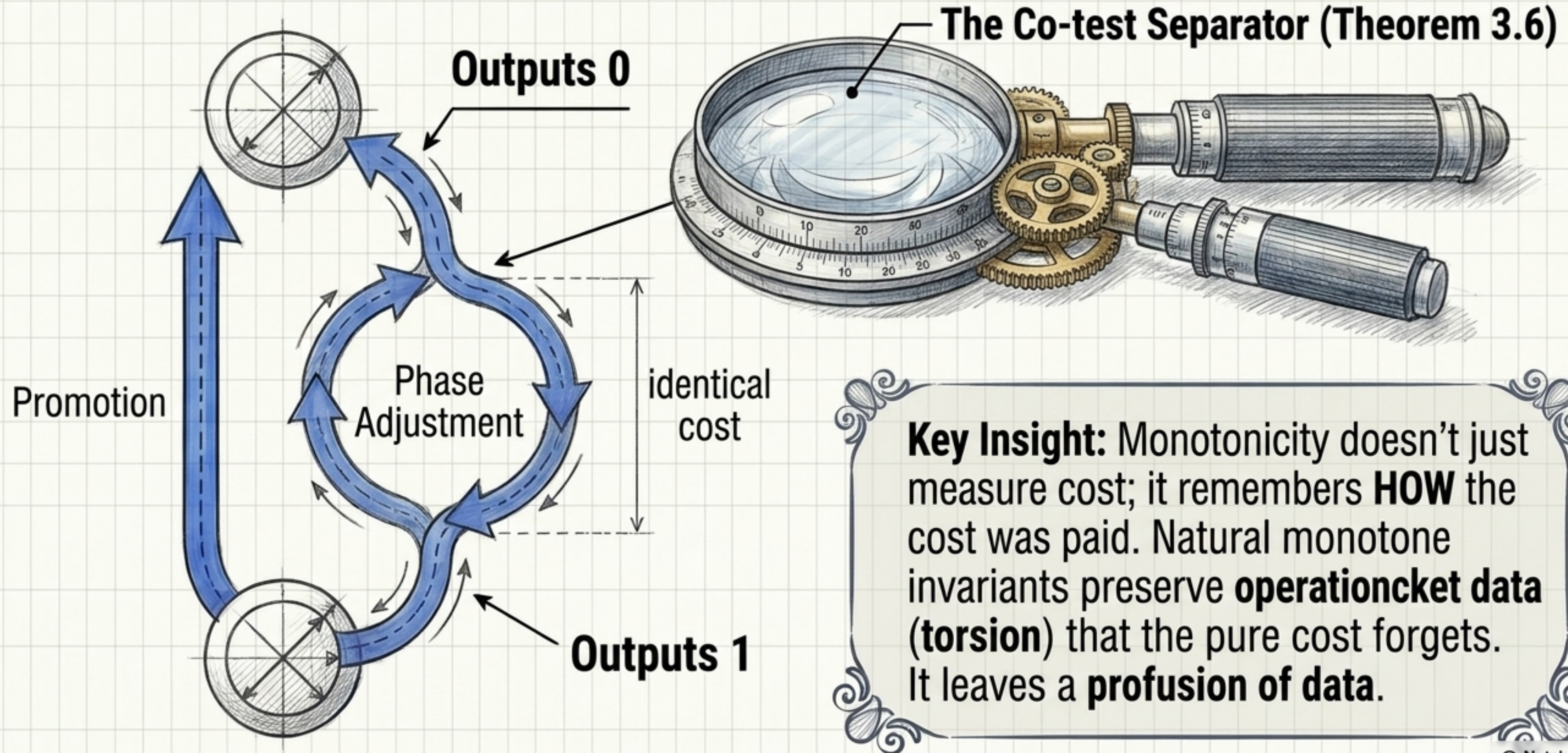
The Goal

In the style of Chentsov, determine which quantities are **actually characterized** by the **invariances** a continuation system supports.

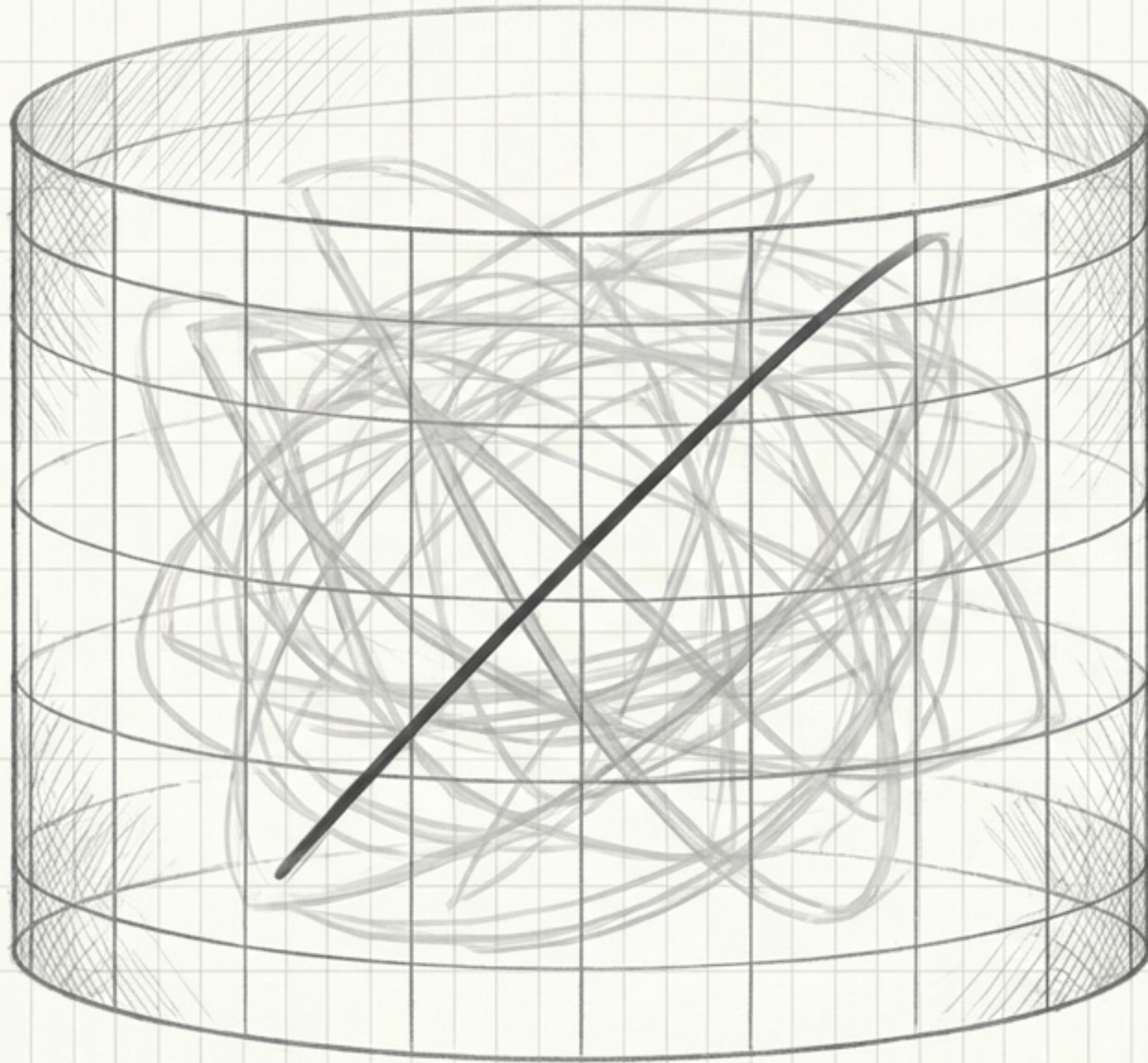
The Filters

- **(M) Monotonicity**: Does it survive under any mapping that respects operations and non-increasing costs?
- **(I) Sufficiency**: Does it hold perfectly equal under lossless, repair-sufficient mappings?

Monotonicity alone is maximally underdetermined



The missing key: Repair sufficiency is provably scarce



What is sufficiency?

A mapping that coarse-grains a system without erasing any repair-relevant distinction (an isometric embedding).

The Scarcity

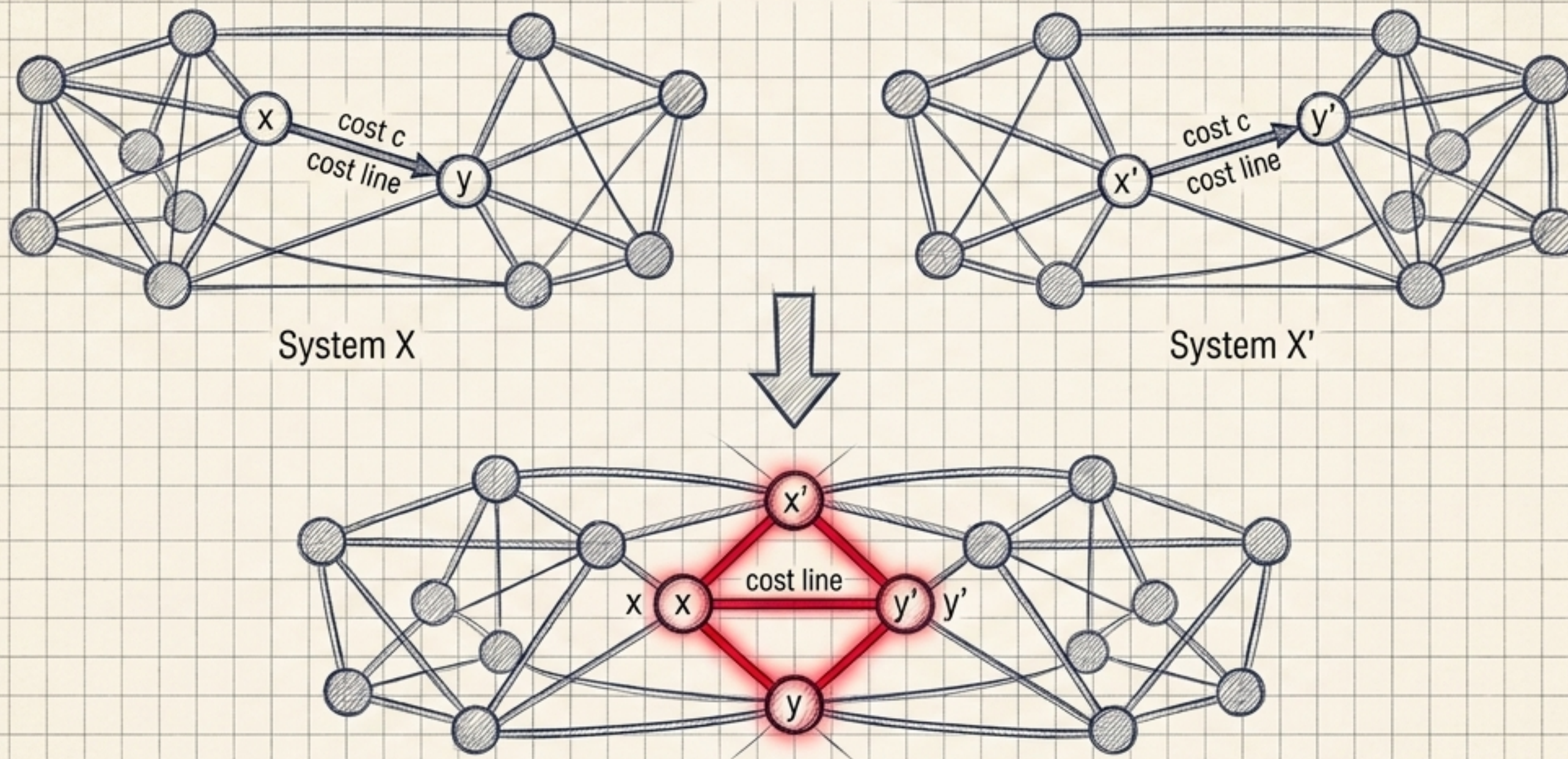
Chentsov's classical proof relies on an abundance of sufficient statistics. In our finite, irreversible continuation system, only 1 of the 22 endomorphisms (the identity) is E-sufficient.

The Problem

We cannot collapse the invariant using only the internal structure of one system.

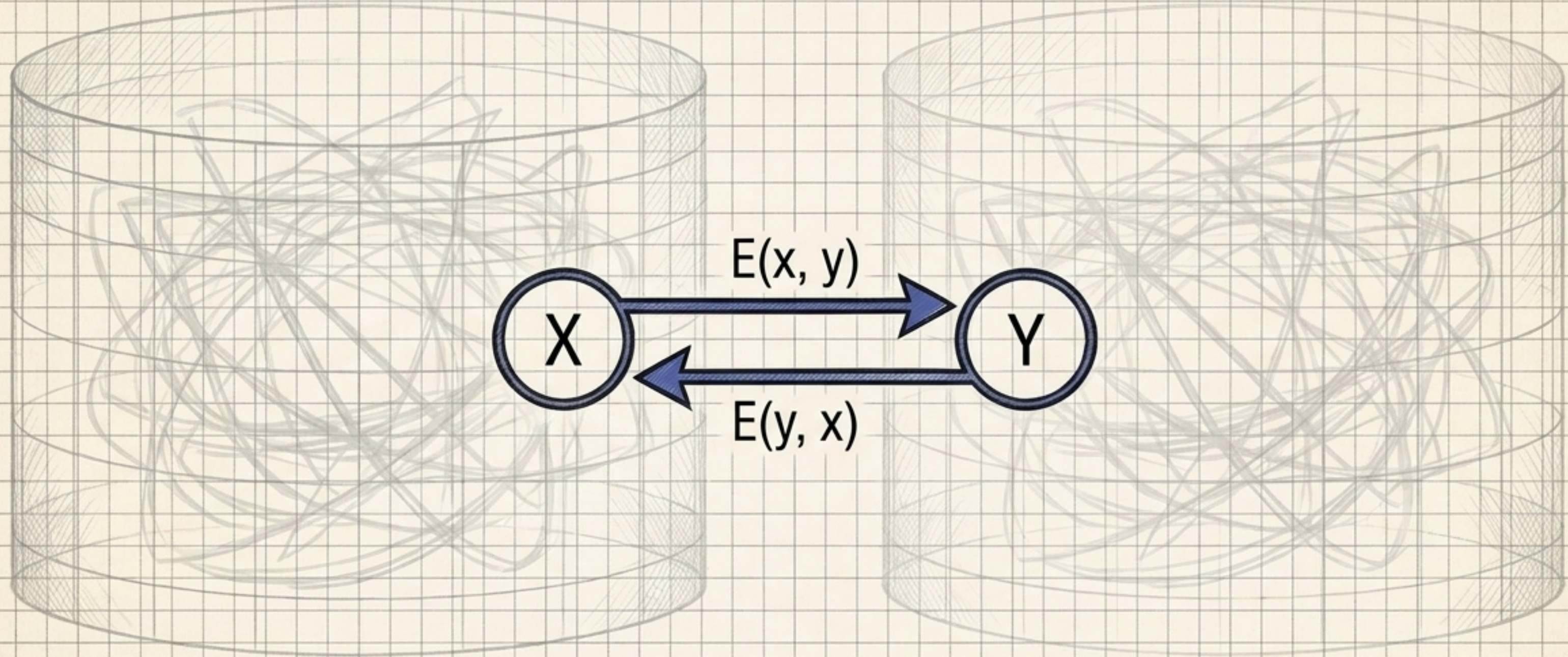
The Hack: Manufacturing sufficiency through Amalgamation

Lemma 3.8



The Result: By gluing two systems along a pair of states with matching costs, both inclusions into the new union system become E-sufficient admissibility morphisms. Sufficiency has been forced into existence between systems.

The Collapse: All that survives is the bidirected cost



Theorem 3.9: Adding invariance under repair-sufficient morphisms cuts the profusion down exactly. What survives every distinction-preserving redescription of a continuation system is just two numbers per ordered pair: the cost of getting there, and the cost of getting back. Directedness survives; operational torsion does not.

The Invariance Filter: What survives lossless recoding?

Monotonicity (M)

Monotonicity + Sufficiency (M)+(I)

What it sees: The pair (Enrichment, Operational Structure).

What it sees: The Bidirected Cost alone.

Locus of Rule: Works entirely within any single system.

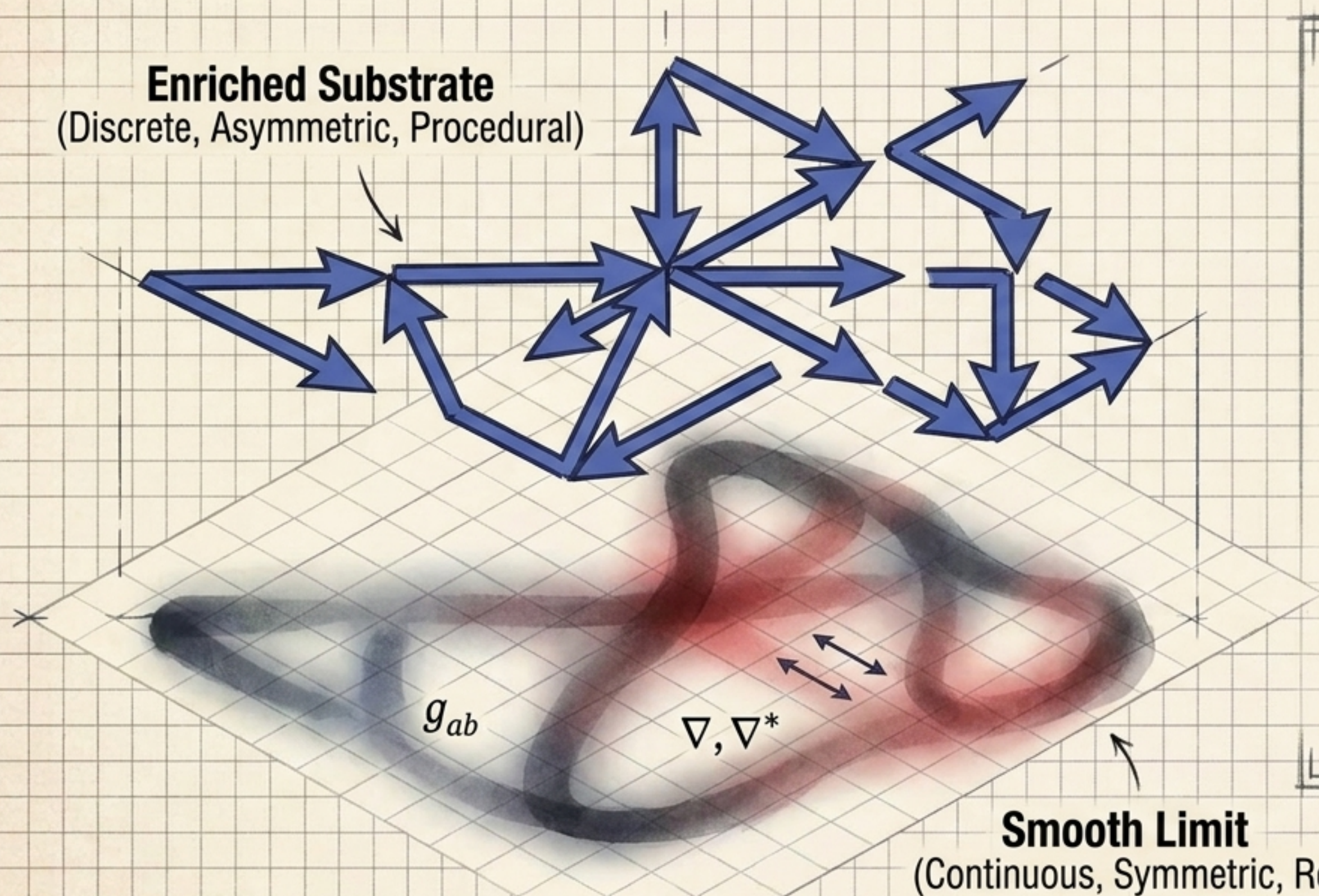
Locus of Rule: Requires gluing across multiple systems (Amalgamation).

Sensitivity to Torsion: High. It remembers if a move was a promotion or a phase shift.

Sensitivity to Torsion: Erased. All operational-bracket data is quotiented away.

Conclusion: The operational stratum is real, but perspective-bound. It is the content of a description, while the bidirected cost is the invariant content of the system.

Dual geometry is a tangent approximation of the enriched substrate



The Smooth Limit (Simplex Relaxation)

- When finite states are relaxed into probability mixtures, smooth contrast functions emerge.
- The symmetric metric and dual connections of information geometry arise merely as the second- and third-order jets of this smooth contrast.
- Continuous geometry is not the foundation; it is a shadow cast by **the discrete categorical reality.**

Repair generates the geometry; it does not live inside it.

1. **The Discrete Reality:** The finite theory of continuation is enriched-categorical, not differential-geometric.
2. **The True Invariant:** The sufficiency-invariant content of a finite continuation system is exactly its bidirected executable cost.
3. **The Shadow:** What is derived from first principles is not the covariant derivative of general relativity, but the directed, asymmetric thing of which it is merely a smooth, reversible shadow.

*To speak invariantly about a continuation system is to speak of just two things:
what it costs to arrive, and what it costs to return.*